

Quadratic Equations

What a quadratic equation is, how to find real roots by factorisation and the quadratic formula, and how the discriminant tells you the nature of the roots.

Introduction

You have already worked with linear equations like $2x + 3 = 0$, which have only x to the power 1. In this chapter we move one step up, to equations where the highest power of x is 2. These are called quadratic equations, and they appear in many real situations, from finding the sides of a rectangle when only the area and perimeter are known, to problems about time, speed and height.

The chapter has two main goals: first, to know when an equation is quadratic and what its roots mean; second, to learn two systematic ways to find those roots and a quick test to tell what kind of roots you will get without solving fully.

What is a Quadratic Equation?

A quadratic equation in one variable is an equation that can be written in the form

Standard Form of a Quadratic Equation

$$ax^2 + bx + c = 0$$

where a , b and c are real numbers and **$a \neq 0$** . Here a is the coefficient of x^2 , b is the coefficient of x , and c is the constant term.

The condition $a \neq 0$ is what makes it quadratic. If a were 0, the x^2 term would disappear and you would be left with a linear equation.

Not every equation that looks quadratic at first is actually quadratic, you need to simplify it first and check the power of x .

Worked Example

Check whether the following are quadratic equations:

(i) $x(x + 1) + 8 = (x + 2)(x - 2)$ (ii) $(x + 2)^3 = x^3 - 4$

(i) Left side: $x^2 + x + 8$. Right side: $x^2 - 4$. Bringing all terms to one side gives $x^2 + x + 8 - x^2 + 4 = 0$, which simplifies to $x + 12 = 0$. Since the x^2 term cancels, this is **not** a quadratic equation.

(ii) Expanding $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$. So $x^3 + 6x^2 + 12x + 8 = x^3 - 4$, which simplifies to $6x^2 + 12x + 12 = 0$, or $x^2 + 2x + 2 = 0$. The highest power is 2 and $a \neq 0$, so this **is** a quadratic equation.

Roots vs Zeroes

These two words sound similar and textbooks sometimes use them interchangeably, but it helps to keep the distinction clear.

Root of a Quadratic Equation

A real number α is called a **root** (or solution) of $ax^2 + bx + c = 0$ if it satisfies the equation, that is, if $a\alpha^2 + b\alpha + c = 0$.

A **zero of the polynomial** $p(x) = ax^2 + bx + c$ is a number α for which $p(\alpha) = 0$. For a quadratic polynomial, the zeroes of the polynomial and the roots of the corresponding equation $p(x) = 0$ are the same numbers.

In short, "zero" belongs to the polynomial $p(x)$, and "root" belongs to the equation $p(x) = 0$, but they refer to the same values. A quadratic equation has at most two roots, just as a quadratic polynomial has at most two zeroes. We can check a possible root simply by substituting it.

Worked Example

Check whether $x = 2$ and $x = 3$ are roots of $x^2 - 5x + 6 = 0$.

For $x = 2$: $(2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$, so $x = 2$ **is** a root.

For $x = 3$: $(3)^2 - 5(3) + 6 = 9 - 15 + 6 = 0$, so $x = 3$ **is** also a root. This equation has two roots, 2 and 3.

Solving by Factorisation

Factorisation is usually the fastest method when the quadratic can be split neatly. The idea is to write the quadratic as a product of two linear factors and then use the fact that if a product is zero, at least one factor must be zero.

For $ax^2 + bx + c = 0$, we look for two numbers whose product is $a \times c$ and whose sum is b . Then we split the middle term bx using those two numbers, factor by grouping, and solve.

Worked Example

Solve $x^2 - 7x + 12 = 0$ by factorisation.

Here $a \times c = 1 \times 12 = 12$ and $b = -7$. We need two numbers with product 12 and sum -7 . They are -3 and -4 .

$$x^2 - 3x - 4x + 12 = 0$$

$$x(x - 3) - 4(x - 3) = 0 \implies (x - 3)(x - 4) = 0$$

So $x - 3 = 0$ or $x - 4 = 0$, which gives **$x = 3$ or $x = 4$** .

Factorisation works well when such factors are easy to identify. When they are not, the quadratic formula is the reliable alternative.

The Quadratic Formula

For $ax^2 + bx + c = 0$, with $a \neq 0$, the quadratic formula gives the roots directly.

Quadratic Formula

For $ax^2 + bx + c = 0$, where $a \neq 0$, the roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is sometimes called Sridharacharya's formula. You just identify a , b and c from the equation (be careful with signs), substitute them in, and simplify. The $\pm$ means you get two values, one with $+$ and one with $-$, unless the square root part is zero.

Worked Example

Solve $2x^2 - 7x + 3 = 0$ using the quadratic formula.

Here $a = 2$, $b = -7$, $c = 3$.

$$b^2 - 4ac = (-7)^2 - 4(2)(3) = 49 - 24 = 25$$

$$x = \frac{-(-7) \pm \sqrt{25}}{2(2)} = \frac{7 \pm 5}{4}$$

So $x = (7 + 5)/4 = 12/4 = 3$, or $x = (7 - 5)/4 = 2/4 = 1/2$. So **$x = 3$ or $x = 1/2$** .

The formula is especially useful when factorisation is not obvious.

Discriminant and Nature of Roots

You do not always need to find the actual roots, sometimes you only need to know what kind of roots an equation has. That information is hidden in the expression under the square root in the quadratic formula.

Discriminant

For $ax^2 + bx + c = 0$, the **discriminant** is

$$D = b^2 - 4ac$$

It determines the nature of the roots:

1. If $D > 0$: there are two distinct real roots. They are $[-b + \sqrt{D}]/2a$ and $[-b - \sqrt{D}]/2a$. If D is a perfect square, the roots are rational; otherwise they are irrational but still real.

2. If $D = 0$: there are two equal real roots (one repeated root). Each root is $-b/2a$. The quadratic is a perfect square: $a(x + b/2a)^2 = 0$.

3. If $D < 0$: there are no real roots. The square root of a negative number is not a real number, so the equation has no solution in real numbers (you are not required to work with complex numbers at this stage).

Worked Example

Find the nature of the roots of each equation and, if real, find them:

(i) $3x^2 - 2x + 1/3 = 0$ (ii) $2x^2 - 6x + 3 = 0$ (iii) $x^2 + x + 1 = 0$

(i) $D = (-2)^2 - 4(3)(1/3) = 4 - 4 = 0$. So the equation has **two equal real roots**. Each root is $-b/2a = 2/6 = 1/3$. So $x = 1/3$ (repeated).

(ii) $D = (-6)^2 - 4(2)(3) = 36 - 24 = 12 > 0$, so there are **two distinct real roots**. They are $x = [6 \pm \sqrt{12}]/4 = [6 \pm 2\sqrt{3}]/4 = [3 \pm \sqrt{3}]/2$.

(iii) $D = 1^2 - 4(1)(1) = 1 - 4 = -3 < 0$, so there are **no real roots**.

A common exam question asks: "Find the value of k for which the equation has equal roots." That simply means set $D = 0$ and solve for k .

Summary

- **Quadratic equation:** $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$. Check by simplifying first, the x^2 term must survive.
- **Root:** a number α that satisfies $a\alpha^2 + b\alpha + c = 0$. The roots of $p(x) = 0$ are the same as the zeroes of the polynomial $p(x)$. A quadratic has at most two roots.
- **Factorisation:** split the middle term using two numbers with product $a \times c$ and sum b , factor by grouping, then set each factor to zero.
- **Quadratic formula:** $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$. Substitute a, b, c with correct signs.
- **Discriminant $D = b^2 - 4ac$:** $D > 0$ gives two distinct real roots, $D = 0$ gives two equal real roots ($-b/2a$), $D < 0$ gives no real roots.

Practice Questions

1. Check whether $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation.

2. Find the roots of $x^2 - 3x - 10 = 0$ by factorisation.

3. Solve $2x^2 + x - 4 = 0$ using the quadratic formula. Give answers in simplest surd form.

4. Find the value of k for which the equation $kx(x - 2) + 6 = 0$ has two equal real roots.

5. Find the discriminant of $3x^2 - 4\sqrt{3}x + 4 = 0$ and state the nature of its roots. Find the roots if they are real.

6. A cottage industry produces a certain number of pottery articles in a day. On a particular day, the cost of production of each article was ₹ 3 more than twice the number of articles produced. If the total cost on that day was ₹ 90, find the number of articles produced.